Chapter 6

Numerical Differentiation and Integration

6.1 INTRODUCTION

In Chapter 3, we were concerned with the general problem of interpolation, viz., given the set of values (x_0, y_0), (x_1, y_1), …, (x_n, y_n) of x and y, to find a polynomial $\phi(x)$ of the lowest degree such that $y(x)$ and $\phi(x)$ agree at the set of tabulated points. In the present chapter, we shall be concerned with the problems of numerical differentiation and integration. That is to say, given the set of values of x and y, as above, we shall derive formulae to compute:

(i) $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \cdots$ for any value of x in $[x_0, x_n]$, and

(ii) $\int_{x_0}^{x_n} y\,dx$.

6.2 NUMERICAL DIFFERENTIATION

The general method for deriving the numerical differentiation formulae is to differentiate the interpolating polynomial. Hence, corresponding to each of the formulae derived in Chapter 3, we may derive a formula for the derivative. We illustrate the derivation with Newton's forward difference formula only, the method of derivation being the same with regard to the other formulae.

Consider Newton's forward difference formula:

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \cdots, \qquad (6.1)$$

where

$$x = x_0 + uh. \qquad (6.2)$$

Then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{1}{h}\left(\Delta y_0 + \frac{2u-1}{2}\Delta^2 y_0 + \frac{3u^2 - 6u + 2}{6}\Delta^3 y_0 + \cdots\right). \qquad (6.3)$$

This formula can be used for computing the value of *dy*/*dx* for *non-tabular values* of *x*. For tabular values of *x*, the formula takes a simpler form, for by setting $x = x_0$ we obtain $u = 0$ from Eq. (6.2), and hence Eq. (6.3) gives

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{1}{h}\left(\Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 - \frac{1}{4}\Delta^4 y_0 + \cdots\right). \qquad (6.4)$$

Differentiating Eq. (6.3) once again, we obtain

$$\frac{d^2y}{dx^2} = \frac{1}{h^2}\left(\Delta^2 y_0 + \frac{6u-6}{6}\Delta^3 y_0 + \frac{12u^2 - 36u + 22}{24}\Delta^4 y_0 + \cdots\right), \qquad (6.5)$$

from which we obtain

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{1}{h^2}\left(\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12}\Delta^4 y_0 + \cdots\right). \qquad (6.6)$$

Formulae for computing higher derivatives may be obtained by successive differentiation. In a similar way, different formulae can be derived by starting with other interpolation formuale. Thus,

(a) Newton's backward difference formula gives

$$\left[\frac{dy}{dx}\right]_{x=x_n} = \frac{1}{h}\left(\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \cdots\right) \qquad (6.7)$$

and

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_n} = \frac{1}{h^2}\left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n + \frac{5}{6}\nabla^5 y_n + \cdots\right). \qquad (6.8)$$

(b) Stirling's formula gives

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{1}{h}\left(\frac{\Delta y_{-1} + \Delta y_0}{2} - \frac{1}{6}\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2} + \frac{1}{30}\frac{\Delta^5 y_{-3} + \Delta^5 y_{-2}}{2} + \cdots\right) \qquad (6.9)$$

and

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{1}{h^2}\left(\Delta^2 y_{-1} - \frac{1}{12}\Delta^4 y_{-2} + \frac{1}{90}\Delta^6 y_{-3} - \cdots\right). \tag{6.10}$$

If a derivative is required near the end of a table, one of the following formulae may be used to obtain better accuracy

$$hy_0' = \left(\Delta - \frac{1}{2}\Delta^2 + \frac{1}{3}\Delta^3 - \frac{1}{4}\Delta^4 + \frac{1}{5}\Delta^5 - \frac{1}{6}\Delta^6 + \cdots\right)y_0 \tag{6.11}$$

$$= \left(\Delta + \frac{1}{2}\Delta^2 - \frac{1}{6}\Delta^3 + \frac{1}{12}\Delta^4 - \frac{1}{20}\Delta^5 + \frac{1}{30}\Delta^6 - \cdots\right)y_{-1} \tag{6.12}$$

$$h^2 y_0'' = \left(\Delta^2 - \Delta^3 + \frac{11}{12}\Delta^4 - \frac{5}{6}\Delta^5 + \frac{137}{180}\Delta^6 - \frac{7}{10}\Delta^7 + \frac{363}{560}\Delta^8 - \cdots\right)y_0 \tag{6.13}$$

$$= \left(\Delta^2 - \frac{1}{12}\Delta^4 + \frac{1}{12}\Delta^5 - \frac{13}{180}\Delta^6 + \frac{11}{180}\Delta^7 - \frac{29}{560}\Delta^8 + \cdots\right)y_{-1} \tag{6.14}$$

$$hy_n' = \left(\nabla + \frac{1}{2}\nabla^2 + \frac{1}{3}\nabla^3 + \frac{1}{4}\nabla^4 + \frac{1}{5}\nabla^5 + \frac{1}{6}\nabla^6 + \frac{1}{7}\nabla^7 + \frac{1}{8}\nabla^8 + \cdots\right)y_n \tag{6.15}$$

$$= \left(\nabla - \frac{1}{2}\nabla^2 - \frac{1}{6}\nabla^3 - \frac{1}{12}\nabla^4 - \frac{1}{20}\nabla^5 - \frac{1}{30}\nabla^6 - \frac{1}{42}\nabla^7 - \frac{1}{56}\nabla^8 - \cdots\right)y_{n+1}$$

(6.16)

$$h^2 y_n'' = \left(\nabla^2 + \nabla^3 + \frac{11}{12}\nabla^4 + \frac{5}{6}\nabla^5 + \frac{137}{180}\nabla^6 + \frac{7}{10}\nabla^7 + \frac{363}{560}\nabla^8 + \cdots\right)y_n \tag{6.17}$$

$$= \left(\nabla^2 - \frac{1}{12}\nabla^4 - \frac{1}{12}\nabla^5 - \frac{13}{180}\nabla^6 - \frac{11}{180}\nabla^7 - \frac{29}{560}\nabla^8 - \cdots\right)y_{n+1}. \tag{6.18}$$

For more details, the reader is referred to Interpolation and Allied Tables. The following examples illustrate the use of the formulae stated above.

Example 6.1 From the following table of values of x and y, obtain dy/dx and d^2y/dx^2 for $x = 1.2$:

x	y	x	y
1.0	2.7183	1.8	6.0496
1.2	3.3201	2.0	7.3891
1.4	4.0552	2.2	9.0250
1.6	4.9530		

The difference table is

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5	Δ^6
1.0	2.7183						
		0.6018					
1.2	3.3201		0.1333				
		0.7351		0.0294			
1.4	4.0552		0.1627		0.0067		
		0.8978		0.0361		0.0013	
1.6	4.9530		0.1988		0.0080		0.0001
		1.0966		0.0441		0.0014	
1.8	6.0496		0.2429		0.0094		
		1.3395		0.0535			
2.0	7.3891		0.2964				
		1.6359					
2.2	9.0250						

Here $x_0 = 1.2,\ y_0 = 3.3201$ and $h = 0.2$. Hence Eq. (6.11) gives

$$\left[\frac{dy}{dx}\right]_{x=1.2} = \frac{1}{0.2}\left[0.7351 - \frac{1}{2}(0.1627) + \frac{1}{3}(0.0361) - \frac{1}{4}(0.0080) + \frac{1}{5}(0.0014)\right]$$

$$= 3.3205.$$

If we use formula (6.12), then we should use the differences diagonally downwards from 0.6018 and this gives

$$\left[\frac{dy}{dx}\right]_{x=1.2} = \frac{1}{0.2}\left[0.6018 + \frac{1}{2}(0.1333) - \frac{1}{6}(0.0294) + \frac{1}{12}(0.0067) - \frac{1}{20}(0.0013\right]$$

$$= 3.3205, \text{ as before.}$$

Similarly, formula (6.13) gives

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.2} = \frac{1}{0.04}\left[0.1627 - 0.0361 + \frac{11}{12}(0.0080) - \frac{5}{6}(0.0014)\right] = 3.318.$$

Using formula (6.14), we obtain

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.2} = \frac{1}{0.04}\left[0.1333 - \frac{1}{12}(0.0067) + \frac{1}{12}(0.0013)\right] = 3.32.$$

Example 6.2 Calculate the first and second derivatives of the function tabulated in the preceding example at the point $x = 2.2$ and also dy/dx at $x = 2.0$.

We use the table of differences of Example 6.1. Here $x_n = 2.2$, $y_n = 9.0250$ and $h = 0.2$. Hence formula (6.15) gives

$$\left[\frac{dy}{dx}\right]_{x=2.2} = \frac{1}{0.2}\left[1.6359 + \frac{1}{2}(0.2964) + \frac{1}{3}(0.0535) + \frac{1}{4}(0.0094) + \frac{1}{5}(0.0014)\right]$$

$$= 9.0228.$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=2.2} = \frac{1}{0.04}\left[0.2964 + 0.0535 + \frac{11}{12}(0.0094) + \frac{5}{6}(0.0014)\right] = 8.992.$$

To find dy/dx at $x = 2.0$, we can use either (6.15) or (6.16). Formula (6.15) gives

$$\left[\frac{dy}{dx}\right]_{x=2.0} = \frac{1}{0.2}\left[1.3395 + \frac{1}{2}(0.2429) + \frac{1}{3}(0.0441) + \frac{1}{4}(0.0080) + \frac{1}{5}(0.0013) + \frac{1}{6}(0.0001)\right]$$

$$= 7.3896.$$

whereas from formula (6.16), we obtain

$$\left[\frac{dy}{dx}\right]_{x=2.0} = \frac{1}{0.2}\left[1.6359 - \frac{1}{2}(0.2964) - \frac{1}{6}(0.0535) - \frac{1}{12}(0.0094) - \frac{1}{20}(0.0014)\right]$$

$$= 7.3896.$$

Example 6.3 Find dy/dx and d^2y/dx^2 at $x = 1.6$ for the tabulated function of Example 6.1.

Choosing $x_0 = 1.6$, formula (6.9) gives

$$\left[\frac{dy}{dx}\right]_{x=1.6} = \frac{1}{0.2}\left(\frac{0.8978 + 1.0966}{2} - \frac{1}{2}\frac{0.0361 + 0.0441}{2} + \frac{1}{30}\frac{0.0013 + 0.0014}{2}\right)$$

$$= 4.9530.$$

Similarly, formula (6.10) yields

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.6} = \frac{1}{0.04}\left[0.1988 - \frac{1}{12}(0.0080) + \frac{1}{90}(0.0001)\right] = 4.9525.$$

In the preceding examples, the tabulated function is e^x and hence it is easy to see that the error is considerably more in the case of the second derivatives. This is due to the reason that although the tabulated function and its approximating polynomial would agree at the set of data points, *their slopes at these points may vary considerably*. Numerical differentiation, is, therefore, an unsatisfactory process and should be used only in 'rare cases.' The next section will be devoted to a discussion of errors in the numerical differentiation formulae.

6.2.1 Errors in Numerical Differentiation

The numerical computation of derivatives involves two types of errors, viz. *truncation errors* and *rounding errors*. These are discussed below.

The truncation error is caused by replacing the tabulated function by means of an interpolating polynomial. This error can usually be estimated by formula (3.7). As noted earlier, this formula is of theoretical interest only, since, in practical computations, we usually do not have any information about the derivative $y^{(n+1)}(\xi)$. However, the truncation error in any numerical differentiation formula can easily be estimated in the following manner. Suppose that the tabulated function is such that its differences of a certain order are small and that the tabulated function is well approximated by the polynomial. (This means that the tabulated function does not have any rapidly varying components.) We know that 2ε is the total absolute error in the values of Δy_i, 4ε in the values of $\Delta^2 y_i$, etc., where ε is the absolute error in the values of y_i. Consider now, for example, Stirling's formula (6.9). This can be written in the form

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{\Delta y_{-1} + \Delta y_0}{2h} + T_1 = \frac{y_1 - y_{-1}}{2h} + T_1, \tag{6.19}$$

where T_1, the truncation error, is given by

$$T_1 = \frac{1}{6h}\left|\frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{2}\right|. \tag{6.20}$$

Similarly, formula (6.10) can be written as

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{1}{h^2}\Delta^2 y_{-1} + T_2, \tag{6.21}$$

where

$$T_2 = \frac{1}{12h^2}|\Delta^4 y_{-2}|. \tag{6.22}$$

The *rounding error*, on the other hand, is inversely proportional to h in the case of first derivatives, inversely proportional to h^2 in the case of second

derivatives, and so on. Thus, *rounding error* increases as h decreases. Considering again Stirling's formula in the form of Eq. (6.19), the rounding error does not exceed $2\varepsilon/2h = \varepsilon/h$, where ε is the maximum error in the value of y_i. On the other hand, the formula

$$\left[\frac{dy}{dx}\right]_{x=x_0} = \frac{\Delta y_{-1} + \Delta y_0}{2h} - \frac{\Delta^3 y_{-2} + \Delta^3 y_{-1}}{12h} + \cdots$$

$$= \frac{y_{-2} - 8y_{-1} + 8y_1 - y_2}{12h} + \cdots \tag{6.23}$$

has the maximum rounding error

$$\frac{18\varepsilon}{12h} = \frac{3\varepsilon}{2h}.$$

Finally, the formula

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_0} = \frac{\Delta^2 y_{-1}}{h^2} + \cdots = \frac{y_{-1} - 2y_0 + y_1}{h^2} + \cdots \tag{6.24}$$

has the maximum rounding error $4\varepsilon/h^2$. It is clear that in the case of higher derivatives, the rounding error increases rather rapidly.

Example 6.4 Assuming that the function values given in the table of Example 6.1 are correct to the accuracy given, estimate the errors in the values of dy/dx and d^2y/dx^2 at $x = 1.6$.

Since the values are correct to 4D, it follows that $\varepsilon < 0.00005 = 0.5 \times 10^{-4}$.

Value of dy/dx at x = 1.6:

$$\text{Truncation error} = \frac{1}{6h}\left|\frac{\Delta^3 y_{-1} + \Delta^3 y_0}{2}\right|, \quad \text{from (6.20)}$$

$$= \frac{1}{6(0.2)}\frac{0.0361 + 0.0441}{2}$$

$$= 0.03342$$

and

$$\text{Rounding error} = \frac{3\varepsilon}{2h}, \quad \text{from (6.23)}$$

$$= \frac{3(0.5)10^{-4}}{0.4}$$

$$= 0.00038.$$

Hence,

$$\text{Total error} = 0.03342 + 0.00038 = 0.0338.$$

Using Stirling's formula from Eq. (6.19), with the first differences, we obtain

$$\left(\frac{dy}{dx}\right)_{x=1.6} = \frac{\Delta y_{-1} + \Delta y_0}{2h} = \frac{0.8978 + 1.0966}{0.4} = \frac{1.9944}{0.4} = 4.9860.$$

The *exact value* is 4.9530 so that the error in the above solution is (4.9860 – 4.9530), i.e., 0.0330, which agrees with the total error obtained above.

Value of d^2y/dx^2 *at* $x = 1.6$: Using Eq. (6.24), we obtain

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.6} = \frac{\Delta^2 y_{-1}}{h^2} = \frac{0.1988}{0.04} = 4.9700$$

so that the error = 4.9700 – 4.9530 = 0.0170.

Also,

$$\textit{Truncation error} = \frac{1}{12h^2}|\Delta^4 y_{-2}| = \frac{1}{12(0.04)} \times 0.0080 = 0.01667$$

and

$$\textit{Rounding error} = \frac{4\varepsilon}{h^2} = \frac{4 \times 0.5 \times 10^{-4}}{0.04} = 0.0050.$$

Hence

$$\text{Total error in } \left[\frac{d^2y}{dx^2}\right]_{x=1.6} = 0.0167 + 0.0050 = 0.0217.$$

6.2.2 Cubic Spline Method

The cubic spline derived in Section 5.2 can conveniently be used to compute the *first* and *second* derivatives of a function. For a natural cubic spline, the recurrence formulae (5.31) or (5.32) may be used to compute the spline second derivatives depending upon the choice of the subdivisions. Then Eq. (5.29) gives the spline in the interval of interest, from which the first derivatives can be computed. For the first derivatives at the tabular points, it would, of course, be easier to use formulae (5.28) and (5.30) directly. If, on the other hand, end conditions involving the first derivatives are given, then recurrence formulae (5.34) or (5.36) may be used to compute the remaining first derivatives.

The following examples illustrate the use of the spline formulae in numerical differentiation.

Example 6.5 We consider the function $y(x) = \sin x$ in $[0, \pi]$.

Here $M_0 = M_N = 0$. Let $N = 2$, i.e., $h = \pi/2$. Then

$$y_0 = y_2 = 0, \quad y_1 = 1 \quad \text{and} \quad M_0 = M_2 = 0.$$

Using formulae (5.32), we obtain

$$M_0 + 4M_1 + M_2 = \frac{6}{h^2}(y_0 - 2y_1 + y_2)$$

or

$$M_1 = -\frac{12}{\pi^2}.$$

Formula (5.29) now gives the spline in each interval. Thus, in $0 \le x \le \pi/2$, we obtain

$$s(x) = \frac{2}{\pi}\left(\frac{-2x^3}{\pi^2} + \frac{3x}{2}\right),$$

which gives

$$s'(x) = \frac{2}{\pi}\left[-\frac{2}{\pi^2}(3x^2) + \frac{3}{2}\right]. \qquad \text{(i)}$$

Hence

$$s'\left(\frac{\pi}{4}\right) = \frac{2}{\pi}\left(-\frac{6}{\pi^2}\frac{\pi^2}{16} + \frac{3}{2}\right) = \frac{9}{4\pi} = 0.71619725.$$

Exact value of $s'(\pi/4) = \cos \pi/4 = 1/\sqrt{2} = 0.70710681.$ The percentage error in the computed value of $s'(\pi/4)$ is 1.28%. From (i),

$$s''(x) = -\frac{24}{\pi^3}x$$

and hence

$$s''\left(\frac{\pi}{4}\right) = -\frac{24}{\pi^3}\frac{\pi}{4} = -\frac{6}{\pi^2} = -0.60792710.$$

Since the exact value is $-1/\sqrt{2},$ the percentage error in this result is 14.03 %. We now consider values of $y = \sin x$ in intervals of 10° from $x = 0$ to $\pi.$ To obtain the spline second derivatives we used a computer and the results are given in the following table (up to $x = 90°$).

	$y''(x)$	
x (in degrees)	*Exact*	*Cubic spline*
10	−0.173 648 178	−0.174 089 426
20	−0.342 020 143	−0.342 889 233
30	−0.500 000 000	−0.501 270 524
40	−0.642 787 610	−0.644 420 964
50	−0.766 044 443	−0.767 990 999
60	−0.866 025 404	−0.868 226 016
70	−0.939 692 621	−0.942 080 425
80	−0.984 807 753	−0.987 310 197
90	−1.000 000 000	−1.002 541 048

It is seen that there is greater inaccuracy in the values of the spline second derivatives.

Example 6.6 From the following data for $y(x)$, find $y'(1.0)$.

x	–2	–1	2	3
$y(x)$	–12	–8	3	5

The function from which the above data was calculated is given by $y = -\frac{1}{15}x^3 - \frac{3}{20}x^2 + \frac{241}{60}x - 3.9$. Hence, the exact value of $y'(1)$ is 3.51667.

To apply the cubic spline formula (5.31), we observe that $h_1 = 1$, $h_2 = 3$ and $h_3 = 1$.

For $i = 1, 2$, the recurrence relation gives:

$$8M_1 + 3M_2 = -2$$

and

$$3M_1 + 8M_2 = -10,$$

since $M_0 = M_3 = 0$. We obtain $M_1 = \frac{14}{55}$ and $M_2 = -\frac{74}{55}$. In $-1 \le x \le 2$, we have

$$s_2(x) = \frac{1}{3}\left[\frac{(2-x)^3}{6}\cdot\frac{14}{55} + \frac{(x+1)^3}{6}\left(-\frac{74}{55}\right)\right]$$

$$+\frac{1}{3}\left[-8-\frac{21}{55}\right](2-x) + \frac{1}{3}\left[3-\frac{9}{6}\left(-\frac{74}{55}\right)\right](x+1)$$

Differentiating the above and putting $x = 1$, we obtain

$$y'(1) \approx s_2'(1.0) = \frac{1}{3}\left[-\frac{7}{55} - \frac{148}{55} + \frac{461}{55} + \frac{276}{55}\right]$$

$$= 3.52727, \text{ on simplification.}$$

6.2.3 Differentiation Formulae with Function Values

In Section 6.2, we developed forward, backward and central difference approximations of derivatives in terms of finite differences. From the computational point of view, it would be convenient to express the numerical differentiation formulae in terms of function values. We list below some differentiation formulae for use in numerical computations.

(i) *Forward Differences*

$$y'(x_i) = \frac{y_{i+1} - y_i}{h}; \qquad y'(x_i) = \frac{-y_{i+2} + 4y_{i+1} - 3y_i}{2h} + O(h^2)$$

$$y''(x_i) = \frac{y_i - 2y_{i+1} + y_{i+2}}{h^2}; \qquad y''(x_i) = \frac{-y_{i+3} + 4y_{i+2} - 5y_{i+1} + 2y_i}{h^2};$$

(ii) *Backward Differences*

$$y'(x_i) = \frac{y_i - y_{i-1}}{h}; \qquad y'(x_i) = \frac{3y_i - 4y_{i-1} + y_{i-2}}{2h};$$

$$y''(x_i) = \frac{y_i - 2y_{i-1} + y_{i-2}}{h^2}; \quad y''(x_i) = \frac{2y_i - 5y_{i-1} + 4y_{i-2} - y_{i-3}}{h^2};$$

(iii) *Central Differences*

$$y'(x_i) = \frac{y_{i+1} - y_{i-1}}{2h}; \qquad y'(x_i) = \frac{-y_{i+2} + 8y_{i+1} - 8y_{i-1} + y_{i-2}}{12h};$$

$$y''(x_i) = \frac{y_{i-1} - 2y_i + y_{i+1}}{h^2};$$

$$y''(x_i) = \frac{-y_{i+2} + 16y_{i+1} - 30y_i + 16y_{i-1} - y_{i-2}}{12h^2}$$

These formulae can be derived by using Taylor series expansion of the functions.

6.3 MAXIMUM AND MINIMUM VALUES OF A TABULATED FUNCTION

It is known that the maximum and minimum values of a function can be found by equating the first derivative to zero and solving for the variable. The same procedure can be applied to determine the maxima and minima of a tabulated function.

Consider Newton's forward difference formula

$$y = y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \cdots$$

Differentiating this with respect to p, we obtain

$$\frac{dy}{dp} = \Delta y_0 + \frac{2p-1}{2}\Delta^2 y_0 + \frac{3p^2 - 3p + 2}{6}\Delta^3 y_0 + \cdots \tag{6.25}$$

For maxima or minima $dy/dp = 0$. Hence, terminating the right-hand side, for simplicity, after the third difference and equating it to zero, we obtain the quadratic for p

$$c_0 + c_1 p + c_2 p^2 = 0, \tag{6.26}$$

where

$$\left.\begin{aligned} c_0 &= \Delta y_0 - \frac{1}{2}\Delta^2 y_0 + \frac{1}{3}\Delta^3 y_0 \\ c_1 &= \Delta^2 y_0 - \Delta^3 y_0 \\ \text{and} \quad c_2 &= \frac{1}{2}\Delta^3 y_0. \end{aligned}\right\} \tag{6.27}$$

Values of x can then be found from the relation $x = x_0 + ph$.

Example 6.7 From the following table, find x, correct to two decimal places, for which y is maximum and find this value of y.

x	y
1.2	0.9320
1.3	0.9636
1.4	0.9855
1.5	0.9975
1.6	0.9996

The table of differences is

x	y	Δ	Δ^2
1.2	0.9320		
		0.0316	
1.3	0.9636		–0.0097
		0.0219	
1.4	0.9855		–0.0099
		0.0120	
1.5	0.9975		–0.0099
		0.0021	
1.6	0.9996		

Let $x_0 = 1.2$. Then formula (6.25), terminated after second differences, gives

$$0 = 0.0316 + \frac{2p-1}{2}(-0.0097)$$

from which we obtain $p = 3.8$. Hence

$$x = x_0 + ph = 1.2 + (3.8)\,(0.1) = 1.58.$$

For this value of x, Newton's backward difference formula at $x_n = 1.6$ gives

$$\begin{aligned} y(1.58) &= 0.9996 - 0.2\,(0.0021) + \frac{-0.2\,(-0.2+1)}{2}\,(-0.0099) \\ &= 0.9996 - 0.0004 + 0.0008 \\ &= 1.0. \end{aligned}$$

6.4 NUMERICAL INTEGRATION

The general problem of numerical integration may be stated as follows. Given a set of data points (x_0, y_0), (x_1, y_1), …, (x_n, y_n) of a function $y = f(x)$, where $f(x)$ is not known explicitly, it is required to compute the value of the definite integral

$$I = \int_a^b y\,dx. \tag{6.28}$$

As in the case of numerical differentiation, one replaces $f(x)$ by an interpolating polynomial $\phi(x)$ and obtains, on integration, an approximate value of the definite integral. Thus, different integration formulae can be obtained depending upon the type of the interpolation formula used. We derive in this section a general formula for numerical integration using Newton's forward difference formula.

Let the interval $[a, b]$ be divided into n equal subintervals such that $a = x_0 < x_1 < x_2 < \cdots x_n = b$. Clearly, $x_n = x_0 + nh$. Hence the integral becomes

$$I = \int_{x_0}^{x_n} y\, dx.$$

Approximating y by Newton's forward difference formula, we obtain

$$I = \int_{x_0}^{x_n} \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \cdots \right] dx.$$

Since $x = x_0 + ph, dx = h\, dp$ and hence the above integral becomes

$$I = h\int_0^n \left[y_0 + p\Delta y_0 + \frac{p(p-1)}{2}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{6}\Delta^3 y_0 + \cdots \right] dp,$$

which gives on simplification

$$\int_{x_0}^{x_n} y\, dx = nh\left[y_0 + \frac{n}{2}\Delta y_0 + \frac{n(2n-3)}{12}\Delta^2 y_0 + \frac{n(n-2)^2}{24}\Delta^3 y_0 + \cdots \right]. \quad (6.29)$$

From this *general formula*, we can obtain different integration formulae by putting n = 1, 2, 3, …, etc. We derive here a few of these formulae but it should be remarked that the trapezoidal and Simpson's 1/3-rules are found to give sufficient accuracy for use in practical problems.

6.4.1 Trapezoidal Rule

Setting n = 1 in the general formula (6.29), all differences higher than the first will become zero and we obtain

$$\int_{x_0}^{x_1} y\, dx = h\left(y_0 + \frac{1}{2}\Delta y_0 \right) = h\left[y_0 + \frac{1}{2}(y_1 - y_0) \right] = \frac{h}{2}(y_0 + y_1). \quad (6.30)$$

For the next interval $[x_1, x_2]$, we deduce similarly

$$\int_{x_1}^{x_2} y\, dx = \frac{h}{2}(y_1 + y_2) \quad (6.31)$$

and so on. For the last interval $[x_{n-1}, x_n]$, we have

$$\int_{x_{n-1}}^{x_n} y\,dx = \frac{h}{2}(y_{n-1} + y_n). \tag{6.32}$$

Combining all these expressions, we obtain the rule

$$\int_{x_0}^{x_n} y\,dx = \frac{h}{2}[y_0 + 2(y_1 + y_2 + \cdots + y_{n-1}) + y_n], \tag{6.33}$$

which is known as the *trapezoidal rule*.

The geometrical significance of this rule is that the curve $y = f(x)$ is replaced by n straight lines joining the points (x_0, y_0) and (x_1, y_1); (x_1, y_1) and (x_2, y_2), …, (x_{n-1}, y_{n-1}) and (x_n, y_n). The area bounded by the curve $y = f(x)$, the ordinates $x = x_0$ and $x = x_n$, and the x-axis is then approximately equivalent to the sum of the areas of the n trapeziums obtained.

The error of the trapezoidal formula can be obtained in the following way. Let $y = f(x)$ be continuous, well-behaved, and possess continuous derivatives in $[x_0, x_n]$. Expanding y in a Taylor's series around $x = x_0$, we obtain

$$\int_{x_0}^{x_1} y\,dx = \int_{x_0}^{x_1} \left[y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2} y_0'' + \cdots \right] dx$$

$$= hy_0 + \frac{h^2}{2} y_0' + \frac{h^3}{6} y_0'' + \cdots \tag{6.34}$$

Similarly,

$$\frac{h}{2}(y_0 + y_1) = \frac{h}{2}\left(y_0 + y_0 + hy_0' + \frac{h^2}{2} y_0'' + \frac{h^3}{6} y_0''' + \cdots \right)$$

$$= hy_0 + \frac{h^2}{2} y_0' + \frac{h^3}{4} y_0'' + \cdots \tag{6.35}$$

From Eqs. (6.34) and (6.35), we obtain

$$\int_{x_0}^{x_1} y\,dx - \frac{h}{2}(y_0 + y_1) = -\frac{1}{12} h^3 y_0'' + \cdots, \tag{6.36}$$

which is the error in the interval $[x_0, x_1]$. Proceeding in a similar manner we obtain the errors in the remaining subintervals, viz., $[x_1, x_2]$, $[x_2, x_3]$, … and $[x_{n-1}, x_n]$. We thus have

$$E = -\frac{1}{12}h^3(y_0'' + y_1'' + \cdots + y_{n-1}''), \tag{6.37}$$

where E is the *total error*. Assuming that $y''(\bar{x})$ is the largest value of the n quantities on the right-hand side of Eq. (6.37), we obtain

$$E = -\frac{1}{12}h^3 n y''(\bar{x}) = -\frac{b-a}{12}h^2 y''(\bar{x}) \tag{6.38}$$

since $nh = b - a$.

6.4.2 Simpson's 1/3-Rule

This rule is obtained by putting $n = 2$ in Eq. (6.29), i.e. by replacing the curve by $n/2$ arcs of second-degree polynomials or parabolas. We have then

$$\int_{x_0}^{x_2} y\,dx = 2h\left(y_0 + \Delta y_0 + \frac{1}{6}\Delta^2 y_0\right) = \frac{h}{3}(y_0 + 4y_1 + y_2).$$

Similarly,

$$\int_{x_2}^{x_4} y\,dx = \frac{h}{3}(y_2 + 4y_3 + y_4)$$

$$\vdots$$

and finally

$$\int_{x_{n-2}}^{x_n} y\,dx = \frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n).$$

Summing up, we obtain

$$\int_{x_0}^{x_n} y\,dx = \frac{h}{3}[y_0 + 4(y_1 + y_3 + y_5 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2}) + y_n], \tag{6.39}$$

which is known as *Simpson's 1/3-rule*, or simply Simpson's rule. It should be noted that this rule requires the division of the whole range into an even number of subintervals of width h.

Following the method outlined in Section 6.4.1, it can be shown that the error in Simpson's rule is given by

$$\int_a^b y\,dx = \frac{h}{3}[y_0 + 4(y_1 + y_3 + y_5 + \cdots + y_{n-1}) + 2(y_2 + y_4 + y_6 + \cdots + y_{n-2}) + y_n]$$

$$= -\frac{b-a}{180}h^4 y^{\text{iv}}(\bar{x}), \tag{6.40}$$

where $y^{\text{iv}}(\bar{x})$ is the largest value of the fourth derivatives.

6.4.3 Simpson's 3/8-Rule

Setting $n = 3$ in Eq. (6.29), we observe that all the differences higher than the third will become zero and we obtain

$$\int_{x_0}^{x_3} y\,dx = 3h\left(y_0 + \frac{3}{2}\Delta y_0 + \frac{3}{4}\Delta^2 y_0 + \frac{1}{8}\Delta^3 y_0\right)$$

$$= 3h\left[y_0 + \frac{3}{2}(y_1 - y_0) + \frac{3}{4}(y_2 - 2y_1 + y_0) + \frac{1}{8}(y_3 - 3y_2 + 3y_1 - y_0)\right]$$

$$= \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + y_3).$$

Similarly

$$\int_{x_3}^{x_6} y\,dx = \frac{3h}{8}(y_3 + 3y_4 + 3y_5 + y_6)$$

and so on. Summing up all these, we obtain

$$\int_{x_0}^{x_n} y\,dx = \frac{3h}{8}[(y_0 + 3y_1 + 3y_2 + y_3) + (y_3 + 3y_4 + 3y_5 + y_6) + \cdots$$

$$+ (y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n)]$$

$$= \frac{3h}{8}(y_0 + 3y_1 + 3y_2 + 2y_3 + 3y_4 + 3y_5 + 2y_6 + \cdots$$

$$+ 2y_{n-3} + 3y_{n-2} + 3y_{n-1} + y_n) \qquad (6.41)$$

This rule, called Simpson's (3/8)-rule, is not so accurate as Simpson's rule, the dominant term in the error of this formula being $-(3/80)\,h^5 y^{iv}(\bar{x})$.

6.4.4 Boole's and Weddle's Rules

If we wish to retain differences up to those of the fourth order, we should integrate between x_0 and x_4 and obtain Boole's formula

$$\int_{x_0}^{x_4} y\,dx = \frac{2h}{45}(7y_0 + 32y_1 + 12y_2 + 32y_3 + 7y_4) \qquad (6.42)$$

The leading term in the error of this formula can be shown to be

$$-\frac{8h^7}{945}y^{vi}(\bar{x}).$$

If, on the other hand, we integrate between x_0 and x_6 retaining differences up to those of the sixth order, we obtain Weddle's rule

$$\int_{x_0}^{x_6} y\,dx = \frac{3h}{10}(y_0 + 5y_1 + y_2 + 6y_3 + y_4 + 5y_5 + y_6), \tag{6.43}$$

the error in which is given by $-(h^7/140)y^{\text{vi}}(\bar{x})$.

These two formulae can also be generalized as in the previous cases. It should, however, be noted that the number of strips will have to be a multiple of four in the case of Boole's rule and a multiple of six for Weddle's rule.

6.4.5 Use of Cubic Splines

If $s(x)$ is the cubic spline in the interval (x_{i-1}, x_i), then we have

$$I = \int_{x_0}^{x_n} y\,dx \approx \sum_{i=1}^{n} \int_{x_{i-1}}^{x_i} s(x)\,dx$$

$$= \sum_{i=1}^{n} \int_{x_{i-1}}^{x_i} \left\{ \frac{1}{6h}[(x_i - x)^3 M_{i-1} + (x - x_{i-1})^3 M_i] \right.$$

$$\left. + \frac{1}{h}(x_i - x)\left(y_{i-1} - \frac{h^2}{6} M_{i-1} \right) + \frac{1}{h}(x - x_{i-1})\left(y_i - \frac{h^2}{6} M_i \right) \right\} dx,$$

using Eq. (5.27). On carrying out the integration and simplifying, we obtain

$$I = \sum_{i=1}^{n} \left[\frac{h}{2}(y_{i-1} + y_i) - \frac{h^3}{24}(M_{i-1} + M_i) \right], \tag{6.44}$$

where M_i, the spline second-derivatives, are calculated from the recurrence relation

$$M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2}(y_{i-1} - 2y_i + y_{i+1}), \qquad i = 1, 2, \ldots, n-1.$$

The use of the cubic spline method is demonstrated in Example 6.13.

6.4.6 Romberg Integration

This method can often be used to improve the approximate results obtained by the finite-difference methods. Its application to the numerical evaluation of definite integrals, for example in the use of trapezoidal rule, can be described, as follows. We consider the definite integral

$$= h\left[\frac{1}{2} + \text{Re}\left\{\frac{e^{(-1+i\omega)h}}{1-e^{(-1+i\omega)h}}\right\}\right]$$

$$= h\left[\frac{1}{2} + \text{Re}\left\{\frac{e^{i\omega h}}{e^{h}-e^{i\omega h}}\right\}\right]$$

$$= \frac{h}{2}\cdot\frac{e^{2h}-1}{1+e^{2h}-2e^{h}\cos\omega h}, \quad \text{on simplification} \tag{6.92}$$

The right side of Eq. (6.92) can be further simplified by using hyperbolic functions.

With $h = 0.1$ and $\omega = 1$, formula (6.92) gives

$$I = \int_0^\infty e^{-x}\cos x\,dx$$

$$= \frac{0.1}{2}\cdot\frac{e^{0.2}-1}{1+e^{0.2}-2e^{0.1}\cos(0.1)} \qquad \left[\frac{1}{1+\omega^2} = \frac{1}{10} = 0.1, \omega = 3\right]$$

$$= \frac{0.1}{2}\cdot\frac{0.221402758}{2.221402758 - 2.199299334}$$

$$= 0.500833622,$$

so that the absolute error in the above result is 0.000833622. Table 6.2 gives the values of the integral given in Eq. (6.91) for different values of ω and $h = 0.1$. For comparison, exact values of the integral are also tabulated.

Table 6.2 Values of the Integral (6.91)

ω	*Exact value*	*Value obtained by using* (6.92)
1.0	0.500 000 000	0.500 833 622
3.0	0.100 000 000	0.100 836 955
6.0	0.027 027 027	0.027 875 426
9.0	0.012 195 121	0.013 063 154
12.0	0.006 896 552	0.007 793 302
20.0	0.002 493 766	0.003 524 142

6.10 NUMERICAL DOUBLE INTEGRATION

Formulae for the evaluation of a double integral can be obtained by repeatedly applying the trapezoidal and Simpson's rules derived in Sections 6.4.1 and 6.4.2. We consider, as an example, the double integral defined by

$$I = \int_{y_j}^{y_{j+1}} \int_{x_i}^{x_{i+1}} f(x, y)\, dx\, dy, \tag{6.93}$$

where

$$x_{i+1} = x_i + h \quad \text{and} \quad y_{j+1} = y_j + k.$$

By the repeated 'application of trapezoidal rule' to Eq. (6.93), we get

$$\begin{aligned} I &= \frac{h}{2} \int_{y_j}^{y_{j+1}} [f(x_i, y) + f(x_{i+1}, y)]\, dy \\ &= \frac{hk}{4} [f(x_i, y_j) + f(x_{i+1}, y_j) + f(x_i, y_{j+1}) + f(x_{i+1}, y_{j+1})] \\ &= \frac{hk}{4} [f_{i,j} + f_{i+1,j} + f_{i,j+1} + f_{i+1,j+1}], \end{aligned} \tag{6.94}$$

where $f_{i,j} = f(x_i, y_j)$, etc.

Similarly, applying Simpson's rule to the integral

$$I = \int_{y_{j-1}}^{y_{j+1}} \int_{x_{i-1}}^{x_{i+1}} f(x, y)\, dx\, dy, \tag{6.95}$$

we obtain

$$\begin{aligned} I &= \frac{h}{3} \int_{y_{j-1}}^{y_{j+1}} [f(x_{i-1}, y) + 4f(x_i, y) + f(x_{i+1}, y)]\, dy \\ &= \frac{hk}{9} [f(x_{i-1}, y_{j-1}) + 4f(x_{i-1}, y_j) + f(x_{i-1}, y_{j+1}) \\ &\quad + 4\{f(x_i, y_{j-1}) + 4f(x_i, y_j) + f(x_i, y_{j+1})\} \\ &\quad + f(x_{i+1}, y_{j-1}) + 4f(x_{i+1}, y_j) + f(x_{i+1}, y_{j+1})] \\ &= \frac{hk}{9} [f_{i-1,j-1} + f_{i-1,j+1} + f_{i+1,j-1} + f_{i+1,j+1} \\ &\quad + 4(f_{i-1,j} + f_{i,j-1} + f_{i,j+1} + f_{i+1,j}) + 16 f_{i,j}]. \end{aligned} \tag{6.96}$$

A numerical example is given here.

Example 6.20 Evaluate

$$I = \int_0^1 \int_0^1 e^{x+y} dx\, dy,$$

using the trapezoidal and Simpson's rules. With $h = k = 0.5$, we have the following table of values of e^{x+y}.

	x		
y	0	0.5	1.0
0	1	1.6487	2.7183
0.5	1.6487	2.7183	4.4817
1.0	2.7183	4.4817	7.3891

Using the 'trapezoidal rule' from Eq. (6.94) repeatedly, we obtain

$$I = \frac{0.25}{4}[1.0 + 4(1.6487) + 6(2.7183) + 4(4.4817) + 7.3891]$$
$$= \frac{12.3050}{4}$$
$$= 3.0762.$$

Using 'Simpson's rule' given in Eq. (6.96) repeatedly, we obtain

$$I = \frac{0.25}{9}[1.0 + 2.7183 + 7.3891 + 2.7183 + 4(1.6487 + 4.4817 + 4.4817 + 1.6487) + 16(2.7183)]$$
$$= \frac{26.59042}{9}$$
$$= 2.9545.$$

The 'exact value of the double integral is 2.9525' and therefore it can be verified that the result given by *Simpson's rule* is about sixty times more accurate than that given by the *trapezoidal rule*.

EXERCISES

6.1 Find $\frac{d}{dx} J_0(x)$ at $x = 0.1$ from the following table:

(0, 1.0), (0.1, 0.9975), (0.2, 0.9900), (0.3, 0.9776), (0.4, 0.9604).

6.2 The following table gives angular displacements θ (in radians) at different times t (seconds):
(0, 0.052), (0.02, 0.105), (0.04, 0.168), (0.06, 0.242), (0.08, 0.327), (0.10, 0.408), (0.12, 0.489).
Calculate the angular velocity at $t = 0.06$.

6.3 From the following values of x and y, find $\frac{dy}{dx}$ at $x = 0.6$.

(0.4, 1.5836), (0.5, 1.7974), (0.6, 2.0442), (0.7, 2.3275), (0.8, 2.6511).

6.4 The distances (x cm) traversed by a particle at different times (t seconds) are given below.

t	0.0	0.1	0.2	0.3	0.4	0.5	0.6
x	3.01	3.16	3.29	3.36	3.40	3.38	3.32

Find the velocity of the particle at $t = 0.3$ seconds.

6.5 From the following values of x and y, find $\frac{dy}{dx}$ when (a) $x = 1$, (b) $x = 3$, (c) $x = 6$ and (d) $\frac{d^2y}{dx^2}$ at $x = 3$.

x	0	1	2	3	4	5	6
y	6.9897	7.4036	7.7815	8.1291	8.4510	8.7506	9.0309

6.6 A rod is rotating in a plane about one of its ends. The angle θ (in radians) at different times t (seconds) are given below.

t	0	0.2	0.4	0.6	0.8	1.0
θ	0.0	0.15	0.50	1.15	2.0	3.20

Find its angular velocity and angular acceleration when $t = 0.6$ seconds.

6.7 Tabulate the function $y = f(x) = x^3 - 10x + 6$ at $x_0 = -0.5$, $x_1 = 1.00$ and $x_2 = 2.0$. Compute its first and second derivatives at $x = 1.00$ using Lagrange's interpolation formula. Compare your results with true values.

6.8 Given the following values of x and y, find $\frac{dy}{dx}$ at $x = 2$:

(0, 2), (2, –2), (3, –1).

6.9 A cubic function $y = f(x)$ satisfies the following data:

x	0	1	3	4
$f(x)$	1	4	40	85

Determine $f(x)$ and hence find $f'(2)$ and $f''(2)$.

6.10 The function $y = 3xe^{-x}$ is tabulated below.

(3, 0.4481), (4, 0.2198), (5, 0.1011).

Find $y'(x)$ at $x = 3$, 4 and 5 and compare your results with the exact values.

6.11 From the following values of x and y, find $\frac{dy}{dx}$ at $x = 2$ using the cubic spline method.

(2, 11), (3, 49), (4, 123)

6.12 From the following values of x and y, determine the value of $\frac{dy}{dx}$ at each of the points by fitting a natural cubic spline through them.

(1, 3), (2, 11), (4, 69), (5, 131).

6.13 Given the values of x and y:

(1.2, 0.9320), (1.3, 0.9636), (1.4, 0.9855), (1.5, 0.9975), (1.6, 0.9996), find x, correct to two decimal places, for which y is maximum and find this value of y.

6.14 If $y = A + Bx + Cx^2$ and y_0, y_1, y_2 are the values of y corresponding to $x = 0$, h and $2h$, respectively, prove that

$$\int_0^{2h} y\,dx = \frac{h}{3}(y_0 + 4y_1 + y_2).$$

6.15 Evaluate

(a) $\int_0^{\pi} x \sin x\,dx$ and (b) $\int_{-2}^{2} \frac{x}{5+2x}\,dx$

using the trapezoidal rule with five ordinates.

6.16 State the trapezoidal rule for finding an approximate area under a given curve. A curve is given by the points (x, y) given below.

(0, 23), (0.5, 19), (1.0, 14), (1.5, 11), (2.0, 12.5), (2.5, 16), (3.0, 19), (3.5, 20), (4.0, 20).

Estimate the area bounded by the curve, the x-axis and the extreme ordinates.

6.17 Write an algorithm to evaluate the integral

$$I = \int_{x_0}^{x_n} y\,dx$$

by the trapezoidal rule with step size h. Given the values of x and $y(x)$;

(0, 0.399), (0.5, 0.352), (1.0, 0.242) (1.5, 0.129), (2.0, 0.054)

find an approximate value of

$$\int_0^2 y(x)\,dx.$$

6.18 Estimate the value of the integral

$$\int_1^3 \frac{1}{x}\,dx$$

by Simpson's rule with 4 strips and 8 strips respectively. Determine the error in each case.

6.19 Evaluate

$$I = \int_0^{\pi/2} \sqrt{\sin x}\; dx$$

using Simpson's $\frac{1}{3}$-rule with $h = \pi/12$.

6.20 Using Simpson's $\frac{1}{3}$-rule with $h = 1$, evaluate the integral

$$I = \int_3^7 x^2 \log x\, dx.$$

6.21 Write an algorithm to evaluate $\int_{x_0}^{x_{2n}} y\,dx$ using Simpson's $\frac{1}{3}$-rule when $y(x)$ is given at $x_0,\ x_0 + h,\ \ldots,\ x_0 + 2nh$. Evaluate

$$\int_0^1 e^{-x^2} \sin x\, dx$$

using Simpson's $\frac{1}{3}$-rule with $h = 0.1$.

6.22 Compute the values of

$$I = \int_0^1 \frac{dx}{1 + x^2}$$

using the trapezoidal rule with $h = 0.5$, 0.25 and 0.125. Then obtain a better estimate using Romberg's method. Compare your results with the true value.

6.23 Determine the maximum error in evaluating the integral

$$I = \int_0^{\pi/2} \sin x\, dx$$

by both the trapezoidal and Simpson's $\frac{1}{3}$-rules using four subintervals.

6.24 Estimate the value of the integral

$$I = \int_0^{1/2} \frac{dx}{\sqrt{x}\,\sqrt{1-x}}$$

using the trapezoidal rule. What is its exact value?

6.25 Derive Simpson's $\frac{3}{8}$-rule,

$$\int_{x_0}^{x_3} y\,dx = \frac{3}{8}h\,(y_0 + 3y_1 + 3y_2 + y_3)$$

Using this rule, evaluate

$$\int_0^1 \frac{1}{1+x}\,dx$$

with $h = \frac{1}{6}$. Evaluate the integral by Simpson's $\frac{1}{3}$-rule and compare the results.

6.26 Using the method of undetermined coefficients, derive the formula

$$\int_0^h y(x)\,dx = \frac{h}{2}(y_0 + y_1) + \frac{h^2}{12}(y_0' - y_1')$$

6.27 The function $y = e^x$ satisfies the following data which is unequally spaced:

(1.00, 2.7183), (1.05, 2.8577), (1.10, 3.0042),
(1.15, 3.1582), (1.25, 3.4903), (1.30, 3.6693).

Evaluate the integral

$$\int_{1.0}^{1.3} y\,dx,$$

as accurately as possible.

6.28 Evaluate

$$\int_0^2 \frac{dx}{x^3 + x + 1}$$

by Simpson's $\frac{1}{3}$-rule with $h = 0.25$.

6.29 Use Euler–Maclaurin formula to evaluate the integral

$$I = \int_1^2 (\cos x + \ln x - e^x)\, dx$$

6.30 Use Euler–Maclaurin formula to sum the series

$$S = 1^3 + 2^3 + 3^3 + \cdots + n^3.$$

6.31 Derive the Gauss integration formula when $n = 2$ and apply it to evaluate the integral

$$\int_{-1}^{1} \frac{1}{1+x^2}\, dx.$$

6.32 Use the three-point Gauss formula to evaluate the integral

$$I = \int_0^1 \frac{1}{1+x}\, dx.$$

Compare this result with that obtained by Simpson's $\frac{1}{3}$-rule with $h = 0.125$.

6.33 Let

$$\int_0^2 x^2 f(x)\, dx = k_0\ f(0) + k_1 f(1) + k_2 f(2).$$

If $f(x)$ is approximated by a quadratic through the points $x = 0$, 1, and 2, then find the values of k_0, k_1 and k_2.

6.34 Using the trapezoidal rule, show that

$$\int_0^\infty e^{-x} \cos \omega x\, dx \approx \frac{h}{2} \frac{\sinh h}{\cosh h - \cos \omega h} \qquad \text{(Einarsson)}$$

6.35 Einarsson [1972] derived the formula

$$\int_0^\infty e^{-x} \cos \omega x\, dx \approx \frac{h \sinh h}{\cosh 2h - \cos 2\omega h} (c_1 \cosh h + c_2 \cos \omega h),$$

where

$$c_1 = 2\left[\frac{1 + \cos^2 \omega h}{\omega^2 h^2} - \frac{\sin 2\omega h}{\omega^3 h^3}\right]$$

and $$c_2 = 4\left[\frac{\sin \omega h}{\omega^3 h^3} - \frac{\cos \omega h}{\omega^2 h^2}\right]$$

use this formula to show that

$$\int_0^\infty e^{-x} \cos x \, dx = 0.500001391$$

6.36 If

$$I = \int_0^{\pi/4} \cos^2 x \, dx,$$

compute

$$I\left(0, \frac{\pi}{4}\right), \; I\left(0, \frac{\pi}{8}\right) \text{ and } I\left(\frac{\pi}{8}, \frac{\pi}{4}\right)$$

6.37 Verify the error estimate given in Eq. (6.67) for Problem 6.36.

6.38 Test the error estimate given in Eq. (6.67) in the evaluation of the integral

$$I = \int_0^{\pi/2} \cos x \, dx$$

6.39 Use the trapezoidal rule to evaluate the double integral

$$\int_{-2}^{2} \int_{0}^{4} (x^2 - xy + y^2) \, dx \, dy.$$

6.40 Use Simpson's $\frac{1}{3}$-rule to evaluate the double integral in Problem 6.39 and compare the results with the exact value.

Answers to Exercises

6.1 –0.0505

6.2 3.975

6.3 2.6444

6.4 0.55

6.5 (a) 0.3931 (b) 0.3341 (c) 0.2706 (d) –0.0256.

6.6 3.73 rad/s; 4.48 rad/s^2

6.7 –5.5; 5.0.

6.8 0

6.9 17; 6

6.10 (a) –0.2831 (b) –0.1735 (c) –0.0639.

6.11 29

6.12 $s_1(x) = 2(x - 1)^3 + 3(2 - x) + 9(x - 1),\ 1 \le x \le 2.$

6.13 1.0

6.15 (a) π, (b) –1.0794

6.16 66.5

6.17 0.475

6.18 (a) 1.1000, (b) 1.0987

6.19 1.1873

6.20 177.4816

6.21 0.2947

6.22 0.7855

6.23 $I_T = 0.9871, \quad I_s = 1.0002$

6.24 1.570858

6.25 0.69319; 0.69317

6.27 0.9513

6.28 0.815

6.29 –4.21667

6.31 1.5

6.32 0.693122

6.33 $k_0 = -\frac{2}{15},\ k_1 = \frac{8}{5},\ \ k_2 = \frac{6}{5}.$

6.36 0.269583

6.37 See Example 6.16.

6.38 Error = 0.00014

6.39 112

6.40 106.6667